

Simulation of string's oscillations with velocity Verlet method

Tomasz Chwiej

1st December 2025

1 Introduction

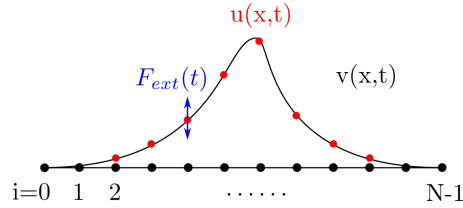


Figure 1: The wavepacket travelling along the string is defined on one-dimensional mesh of nodes, vertical displacement and velocity are described with functions $u(x, t)$ and $v(x, t)$, locally the external time-periodic force $F_{ext}(t)$ may be exerted on string.

The wave equation for an inert (massive) string complemented with frictional term γ and external stimulus $F_{ext}(t)$ reads

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} - \gamma \frac{\partial u}{\partial t} + F_{ext} \sin(\Omega t) \delta_{x, x_p} \quad (1)$$

where $u(x, t)$ is the local vertical displacement of string, $v(x, t) = \frac{\partial u}{\partial t}$ is vertical velocity, δ_{x, x_p} is Kronecker's delta which distincts the point x_p at which the force F_{ext} oscillating with frequency Ω may act on the string, the speed of sound links the tension and linear mass density $c = \sqrt{T/\rho}$. Our aim is to solve this partial differential equation using the finite difference and the velocity Verlet methods.

1.1 Mesh, finite difference method, discretization

Finite difference method requires discretization of time and spatial dimensions. The equidistant nodes are set along the string of length L

$$\Delta x = \frac{L}{N-1}, \quad x_i = \Delta x \cdot i, \quad i = 0, 1, 2, \dots, N-1 \quad (2)$$

$$(3)$$

and the time nodes are

$$t_n = \Delta t \cdot n, \quad n = 0, 1, 2, \dots \quad (4)$$

The time step Δt must be chosen carefully, so as to guarantee the CFL condition is met $|\alpha| = |\frac{c\Delta t}{\Delta x}| \leq 1$, in project we set time step as follows

$$\Delta t = \alpha \frac{\Delta x}{c} \quad (5)$$

From now on all functions are to be defined on mesh $u(x_i, t_n) \rightarrow u_i^n$ and $v(x_i, t_n) \rightarrow v_i^n$, in particular the discretized version of wave equation Eq.1 will be needed

$$a_i^n = c_i^2 \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} - \gamma v_i^n + F_{ext} \sin(\Omega t_n) \delta_{i,i_p} \quad (6)$$

1.2 Boundary conditions (BC)

Two BC will be numerically tested:

- *fixed BC* - both ends of string have fixed positions (Dirichlet)

$$u(0, t) = u(L, t) = 0 \implies u_0^n = u_{N-1}^n = 0 \quad (7)$$

- *open BC* - at both ends the homogeneous Neumann condition is imposed

$$\left. \frac{\partial u}{\partial x} \right|_{x=0, L} = 0 \implies u_0^n \leftarrow u_1^n \text{ and } u_{N-1}^n \leftarrow u_{N-2}^n \quad (8)$$

Remark: as we will impose BC for solution $u(x, t)$ we don't need to define the velocity at these nodes.

1.3 Initial conditions

To start the simulation and get the unique solution we need to specify the initial shape of string $u(x, 0)$ and velocity $v(x, 0)$, both will be defined later.

1.4 Kinetic and potential energies

Without any external force the oscillating string is a Hamilton system with total energy being conserved, while both the kinetic and the potential energy contributions must oscillate as simulation proceeds in time. We calculate the kinetic energy by adding up contributions at all inner nodes

$$E_{kin}(t_n) = \sum_{i=1}^{N-2} \frac{m_i (v_i^n)^2}{2} = \sum_{i=1}^{N-2} \frac{\Delta x \rho_i (v_i^n)^2}{2} \quad (9)$$

and analogically the potential energy

$$E_{pot}(t_n) = \sum_{i=1}^{N-1} \frac{\Delta x T}{2} \left(\left. \frac{\partial u}{\partial x} \right|_{x=x_i} \right)^2 = \sum_{i=1}^{N-1} \frac{\Delta x T}{2} \left(\frac{u_i^n - u_{i-1}^n}{\Delta x} \right)^2 \quad (10)$$

Remark: in Eq.10 we use nonsymmetric backward formula for spatial derivative, hence summation begins at $i = 1$ and ends at last node $i = N - 1$.

1.5 Fundamental modes and frequencies - spectral decomposition of solution

It is well known from basic course in mechanics, that displacement of homogeneous string with immovable ends can be described with sine function. Keeping in mind that all such eigenmodes are solution of linear wave equation, one may describe this fact mathematically expressing the solution as a linear combination of these basis functions

$$u(x, t) = \sum_{k=1}^{\infty} b_k(t) \sin\left(\frac{k\pi x}{L}\right) \quad (11)$$

where we explicitly assume that all time dependencies are holded in coefficients $b_k(t)$. Although, the wave equation is solved on mesh, one might ask which modes contribute mostly to solution. One may naively project the solution on each modes and retrieve coefficients b_k , in this procedure the integral is replaced with summation over the nodes

$$b_k(t) = \frac{2}{L} \int_0^L dx u(x, t) \sin\left(\frac{k\pi x}{L}\right) \quad (12)$$

$$b_k^n = \frac{2}{(N-1)} \sum_{i=1}^{N-2} u_i^n \sin\left(\frac{k\pi i}{N-1}\right) \quad (13)$$

But this is only partly true, and one may convince itself by asking what happens when the string passes through the equilibrium (no displacement along the string lentgh)? The information can not be lost then and must be encoded differently. Because the static string's displacement relates to potential energy, which transforms into kinetic one, the hidden information must be retrieved from the velocity function $v(x, t)$, let's project it on basis functions

$$d_k(t) = \frac{2}{L} \int_0^L dx v(x, t) \sin\left(\frac{k\pi x}{L}\right) \quad (14)$$

$$d_k^n = \frac{2}{(N-1)} \sum_{i=1}^{N-2} v_i^n \sin\left(\frac{k\pi i}{N-1}\right) \quad (15)$$

Coefficients b_k and d_k represent diffrent physical quantities, length and velocity, so they can not be compared directly, both however must enter to equations expressing the potential and kinetic energies. Potential energy

$$E_p(t_n) = \int_0^L dx \frac{T \left(\frac{\partial u}{\partial x}\right)^2}{2} \approx \frac{(N-1)T\Delta x}{4} \sum_{k=1}^{N-1} b_k^n \frac{\sin^2\left(\frac{k\pi}{N-1}\right)}{\Delta x^2} \quad (16)$$

and the kinetic one

$$E_k(t_n) = \sum_{i=1}^{N-1} \frac{m_i(v_i^2)}{2} = \frac{(N-1)\rho\Delta x}{4} \sum_{k=1}^{N-1} d_k^n \quad (17)$$

gives the total energy

$$E_{tot}(t_n) = \sum_{k=1}^{N-1} e_k^n \quad (18)$$

$$e_k^n = \frac{N-1}{4} \left(\frac{T\Delta x \sin^2\left(\frac{k\pi}{N-1}\right)}{\Delta x^2} (b_k^n)^2 + \rho\Delta x (d_k^n)^2 \right) \quad (19)$$

Now, one may easily notice that coefficient e_k^n represents the whole contribution of $k - th$ mode in total energy.

Each $k - th$ fundamental mode has its characteristic frequency of oscillations, for homogeneous mass density distribution these are given by expression

$$\omega_k = \frac{c}{\Delta x} \sqrt{2 \left[1 - \cos\left(\frac{k\pi}{N-1}\right) \right]} \quad (20)$$

1.6 Velocity Verlet method

Now, as we have settled some essential factors needed for Verlet algorithm we are in position to fit this method for the spatially extended object like e.g. considered string

```

initialization:  $N, L, \Delta x, c, T, \alpha, \Delta t, t_{max}, N_t = t_{max}/\Delta t,$ 
                 $\gamma, F_{ext}, K_{modes}$ 

fill arrays:
 $\rho_i = \dots, \quad i = 0, 1, \dots, N-1$ 
 $c_i = \sqrt{\frac{T}{\rho_i}}, \quad i = 0, 1, 2, \dots, N-1$ 

set initial conditions:
 $u_i = \dots, \quad i = 0, 1, \dots, N-1$ 
 $v_i = \dots, \quad i = 0, 1, \dots, N-1$ 

time evolution:
for it=1 to  $N_t$  do

     $t \leftarrow (it-1) \cdot \Delta t$ 

     $a_i \leftarrow c_i^2 \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} - \gamma \cdot v_i + F_{ext} \sin(\Omega t) \cdot \delta_{i,i_p}, \quad i = 1, 2, \dots, N-2$ 

    calculate velocity at half-step:
     $\tilde{v}_i \leftarrow v_i + \frac{\Delta t}{2} a_i, \quad i = 1, 2, \dots, N-2$ 

    calculate new displacements:
     $u_i \leftarrow u_i + \tilde{v}_i \Delta t$ 

    implement BC for new solution:
     $u_0 \leftarrow 0$  and  $u_{N-1} \leftarrow 0$  for fixed BC
    OR
     $u_0 \leftarrow u_1$  and  $u_{N-1} \leftarrow u_{N-2}$  for open BC

    calculate acceleration for new solution and time  $t + \Delta t$ :
     $a_i \leftarrow c_i^2 \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} - \gamma \cdot \tilde{v}_i + F_{ext} \sin(\Omega(t + \Delta t)) \cdot \delta_{i,i_p}, \quad i = 1, 2, \dots, N-2$ 
    calculate new velocities:
     $v_i \leftarrow \tilde{v}_i + a_i \frac{\Delta t}{2}, \quad i = 1, 2, \dots, N-2$ 

    find contributions of a few ( $K_{modes}$ ) fundamental modes:
     $b_k \leftarrow Eq.13, \quad k = 1, 2, \dots, K_{modes}$ 
     $d_k \leftarrow Eq.15, \quad k = 1, 2, \dots, K_{modes}$ 
     $e_k \leftarrow Eq.19, \quad k = 1, 2, \dots, K_{modes}$ 

    calculate other quantities:
     $E_{kin} \leftarrow Eq.9$ 
     $E_{pot} \leftarrow Eq.10$ 

    send output:  $t, E_{kin}, E_{pot}, E_{kin} + E_{pot}$ 
                 $u_i, \quad i = 0, 1, 2, \dots, N-1$ 

```

end do

2 Practical part

Implement Verlet method given in Sec.1.6 on computer and use the following parameters (some of them will be changed later)

- length of string: $L = 1$
- homogeneous linear mass density: $Q = 1$
- tension: $T = 100$
- speed of sound: $c = \sqrt{\frac{T}{Q}}$
- number of nodes: $N = 301$
- spatial step: $\Delta x = \frac{L}{N-1}$
- CFL factor: $\alpha = \frac{1}{2}$
- time step: $\Delta t = \frac{\alpha \Delta x}{c}$
- number of time steps: $N_t = 2000$
- number of fundamental modes: $K_{modes} = 0$

Use initial condition ($t = 0$) for displacement

$$u(x, t) = u_{init} \exp \left[-\frac{(x - x_A - v_{init} \cdot t)^2}{2\sigma^2} \right] \quad (21)$$

and for velocity

$$v(x, 0) = \left. \frac{\partial u}{\partial t} \right|_{t=0} = \frac{v_{init}(x - x_A)}{\sigma^2} u(x, 0) \quad (22)$$

values of parameters: u_{init} , x_A , v_{init} , σ are given below.

Tasks to do:

1. (*fixed BC*, zero initial velocity, homogeneous string)
Assume parameters: $\gamma = 0$, $F_{ext} = 0$, $\Omega = 0$, $u_{init} = 0.01$, $x_A = L/2$, $v_{init} = 0$, $\sigma = L/20$ and *fixed BC*: $u_0^n = u_{N-1}^n = 0$. Carry out simulation then make plot of kinetic, potential and total energies in one figure, make 2D map of string's displacement $u(x, t)$.
2. (*open BC*, zero initial velocity, homogeneous string)
Use the same parameters as in task 1 but now use *open BC*: $u_0^n = u_1^n$ and $u_{N-1}^n = u_{N-2}^n$. Carry out simulation then make plot of kinetic, potential and total energies in one figure, make 2D map of string's displacement $u(x, t)$.
3. (*open BC*, non-zero initial velocity, homogeneous string)
Carry out simulations for parameters and *open BC* as in task 2 but use non-zero initial velocity: (a) $v_{init} = c/2$ and (b) $v_{init} = c$. In both cases plot the kinetic, potential and total energies in one figure, make 2D maps of string's displacement $u(x, t)$.

4. (*open BC*, non-zero initial velocity, non-homogeneous string)

Use the same parameters as in previous task but with the following changes: $v_{init} = c$, $x_A = L/4$ and nonhomogeneous mass density distribution (**speed c_i along the string will also change**)

$$\rho_i = \begin{cases} Q, & x_i \leq \frac{L}{2} \\ 10Q, & x_i > \frac{L}{2} \end{cases} \quad (23)$$

Plot the kinetic, potential and total energies in one figure, make 2D map of string's displacement $u(x, t)$. What happens at the interface between light and massive parts of string?

5. (*fixed BC*, zero initial velocity, homogeneous string, friction, spectral decomposition)

Use parameters: $\gamma = 150$, $v_{init} = 0$, $\rho_i = Q$, $u_{init} = 0.01$, $K_{modes} = 3$ and *fixed BC*. For initial displacement we take a combination of three lowest fundamental modes of string

$$u(x, 0) = u_{init} \left[\sin\left(\frac{1 \cdot \pi x}{L}\right) + \sin\left(\frac{2 \cdot \pi x}{L}\right) + \sin\left(\frac{3 \cdot \pi x}{L}\right) \right] \quad (24)$$

In the course of simulation find the coefficients b_k^n , d_k^n and e_k^n for these three modes and calculate the potential, kinetic and total energies. Plot energies in one figure, plot the coefficients d_k^n and total energy in another single figure, draw also the solution $u(x, t)$ as a map.

6. (*fixed BC*, zero initial velocity, homogeneous string, spectral decomposition, mechanical resonance)

Use *fixed BC*, $v_{init} = 0$, $u_{init} = 0$, $\rho_i = Q$, $\gamma = 0$, $F_{ext} = 100$, $K_{modes} = 3$, $N_t = 10^4$ and following frequencies Ω of external force and points at which it acts on string (i_p)

(a) $\Omega = \omega_1$, $i_p = 30$

(b) $\Omega = \omega_2$, $i_p = 30$

(c) $\Omega = \omega_2$, $i_p = \frac{N-1}{2}$

Use Eq.20 in calculations of values of ω_k .

In the course of each simulation (a,b,c) find the coefficients b_k^n , d_k^n and e_k^n for lowest three modes ($K_{modes} = 3$) and calculate the energies. Plot energies in one figure, plot the coefficients d_k^n and total energy in another single figure, draw also the solution $u(x, t)$ as a map. Compare results and explain the differences observed in (c) in relation to (a) and (b).

3 Example results

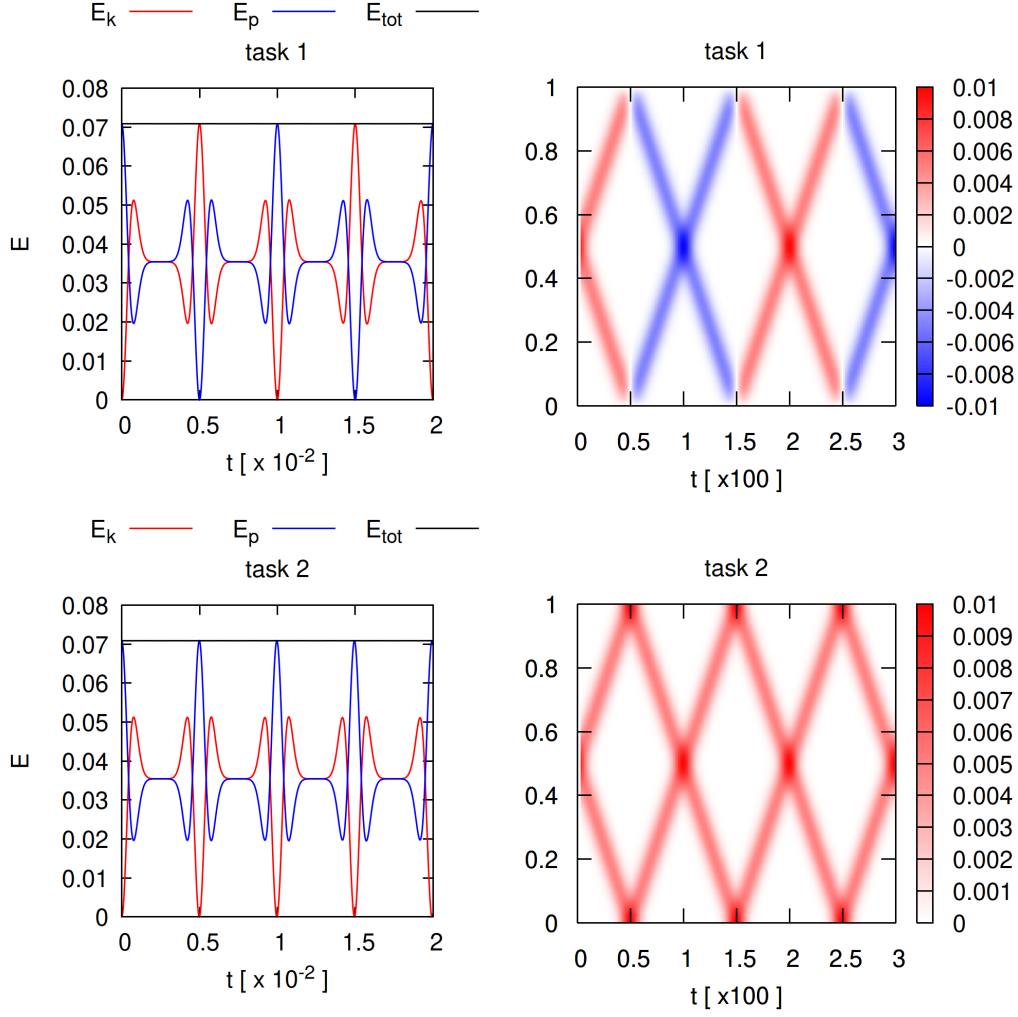


Figure 2: **Tasks 1-2.** Kinetic, potential and total energies (left column) and string displacement $u(x, t)$ (right column) for two boundary conditions: fixed (top row) and open (bottom row). Notice phase reverse in first case, it is absent in second one.

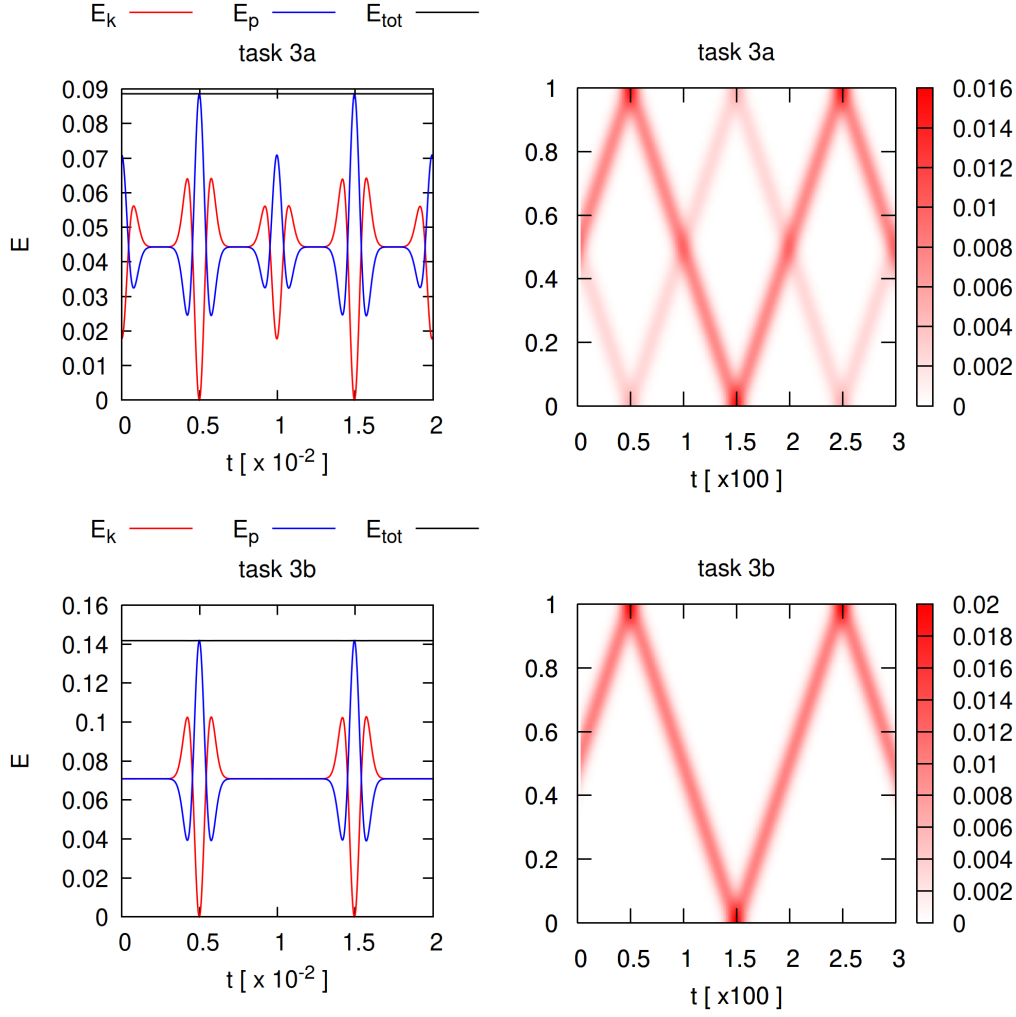


Figure 3: **Task 3.** Kinetic, potential and total energies (left column) and string displacement $u(x,t)$ (right column) for nonzero initial velocity: $v_{init} = 0.5c$ (top row) and $v_{init}=c$ (bottom row). If the group velocity is equal the speed of sound (c) then the whole wavepacket moves in particular direction (bottom), otherwise it divides into two parts (top).

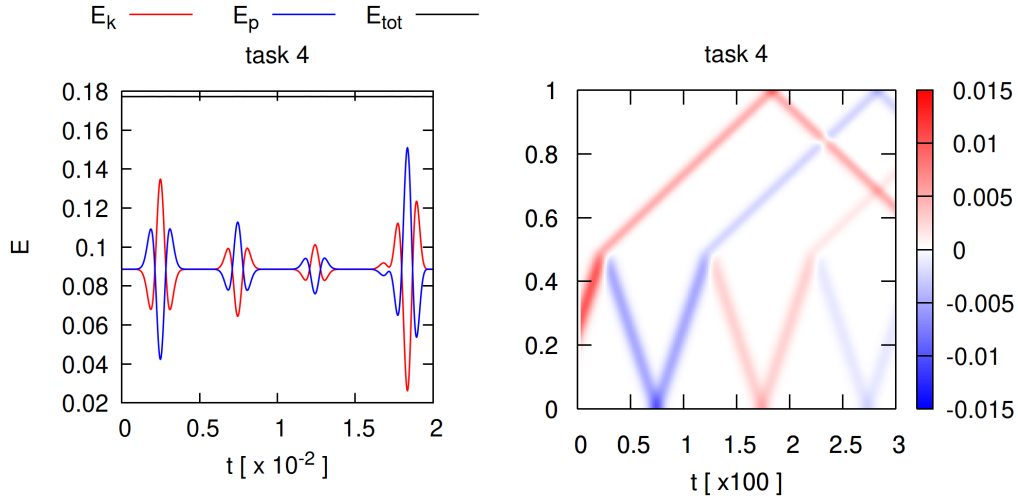


Figure 4: **Task 4.** Kinetic, potential and total energies (left column) and string displacement $u(x,t)$ (right column) for nonhomogeneously distributed mass along the string, for $x < L/2$ linear density is Q while in the second part it is 10 times larger. The wave starting at lighter part moves towards the interface where it is partially reflected, in massive part of string the wave has different velocity.

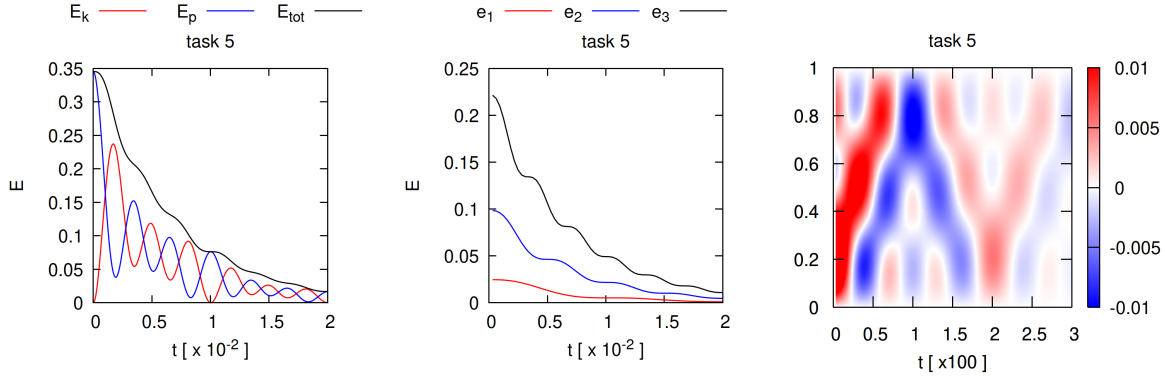


Figure 5: **Task 5.** Kinetic, potential and total energies (left column), the energy contributions of 3 lowest modes (middle) and string displacement $u(x,t)$ (right column) for friction taken in account in simulation. Energy is dissipated nonmonotonically. The string's vertical displacement $u(x,t)$ decreases with time.

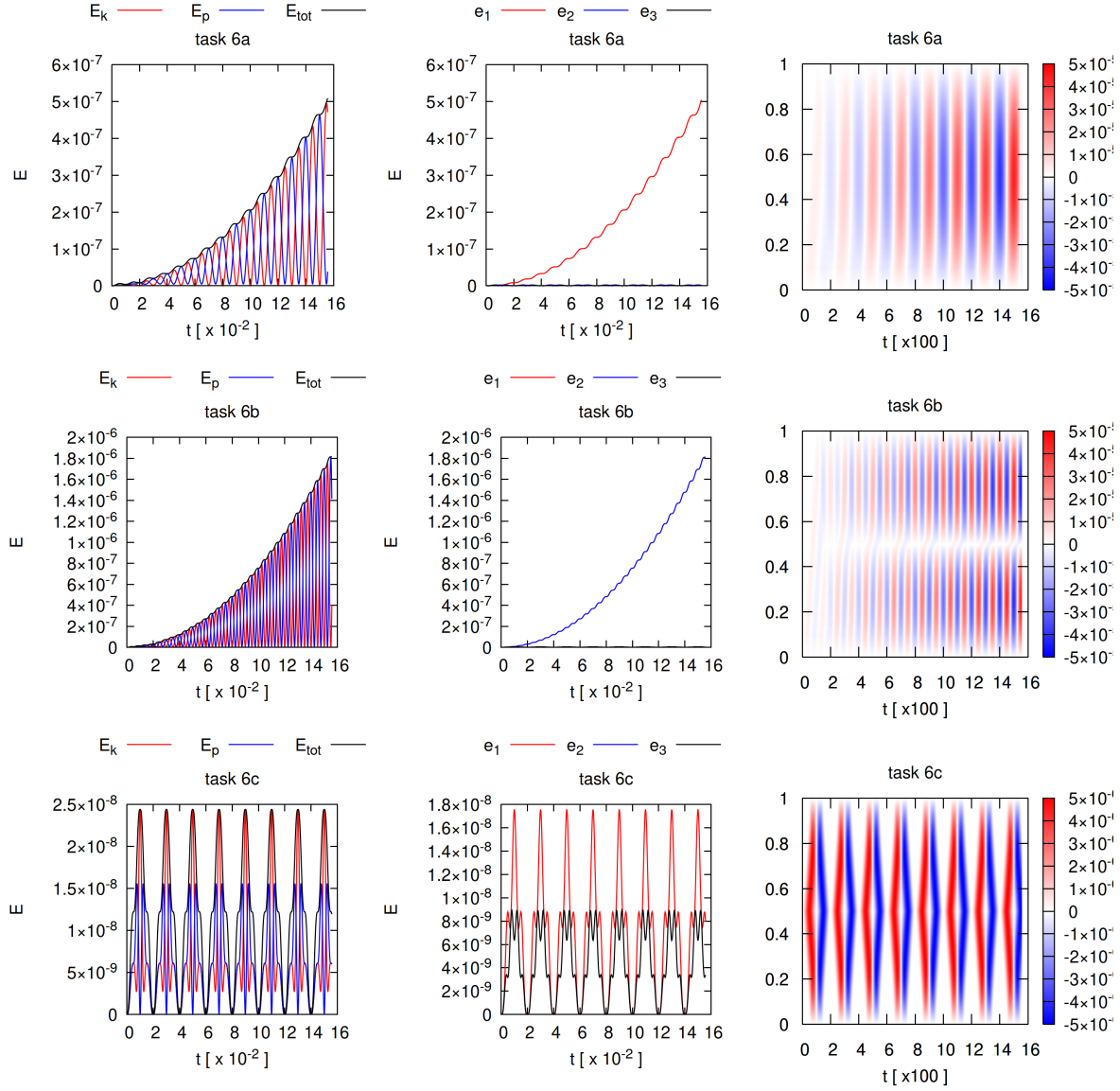


Figure 6: **Task 6.** Kinetic, potential and total energies (left column), the energy contributions of 3 lowest modes (middle) and string displacement $u(x, t)$ (right column) obtained for periodic stimulation of string. (TOP) $i_p = 30$ and $\Omega = \omega_1$; (MIDDLE) $i_p = 30$ and $\Omega = \omega_2$; (BOTTOM) $i_p = (N - 1)/2$ and $\Omega = \omega_2$. In first two cases one of the two lowest modes is excited selectively leaving other modes in rest, in third case the point at which the force is exerted to string lies directly at the node of second mode, for this reason it cannot absorb energy, other modes are excited and fade out periodically but with low amplitudes.